

Chapter 2: Non-Parametric Procedures

Modelling Survival Data in Medical Research

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Part 2

1. Standard Errors and Confidence Intervals
2. Estimating $H(t)$ and $h(t)$
3. Median and Percentiles of Survival Times
4. Q&A

Why Do We Need Standard Errors? (1/2)

The core problem

Our KM estimate $\hat{S}(t)$ is computed from a **sample**.

A different sample from the same population would give a slightly different $\hat{S}(t)$.

The **standard error** measures how much the estimate varies across samples.

Everyday analogy

You survey 100 students: average study time is 4.2 hours.

Survey 100 *different* students: you might get 4.1 or 4.3.

The standard error quantifies this typical variation.

Why Do We Need Standard Errors? (2/2)

Why standard errors grow at later times

At early times: many individuals are still at risk $\Rightarrow$ reliable estimates.

At late times: very *few* individuals remain $\Rightarrow$ much less reliable.

An estimate based on 3 people is far less reliable than one based on 15.

IUD data

At $t = 10$ weeks: 18 women at risk — $\hat{S}(10)$ is reliable.

At $t = 107$ weeks: only 3 women at risk — $\hat{S}(107)$ has a much larger standard error.

This is why the confidence band *widens* as time increases on a KM plot.

Greenwood's Formula

Standard error of $\hat{S}(t)$ at time $t_{(k)} \leq t < t_{(k+1)}$

$$\widehat{\text{se}}\{\hat{S}(t)\} \approx \hat{S}(t) \left[\sum_{j=1}^k \frac{d_j}{n_j(n_j - d_j)} \right]^{1/2}$$

Named after **Major Greenwood** (1926), who first derived this formula.

How to read this formula

- ▶ The sum $\sum d_j/[n_j(n_j - d_j)]$ grows as we include more death times
- ▶ Multiplied by $\hat{S}(t)$, which shrinks over time
- ▶ The standard error **increases** at later times because fewer individuals remain

Greenwood's formula is the standard approach. In R: `survfit()` computes it automatically.

Building a Confidence Interval for $S(t)$

Pointwise 95% confidence interval (linear scale)

$$\hat{S}(t) \pm 1.96 \times \widehat{\text{se}}\{\hat{S}(t)\}$$

The 1.96 comes from the standard normal: $P(-1.96 \leq Z \leq 1.96) = 0.95$.
We have 95% confidence that the *true* $S(t)$ lies in this interval.

Watch out: the interval can go outside $(0, 1)$!

When $\hat{S}(t)$ is close to 0 or 1, the formula can give limits below 0 or above 1.
Fix: replace any limit > 1 with 1.0, and any limit < 0 with 0.0.

Better CI: Complementary Log-Log Transformation

Why transform?

The linear CI has a fundamental flaw: it assumes $\hat{S}(t)$ is normally distributed, which it is not when $\hat{S}(t)$ is close to 0 or 1. A transformation fixes this.

Complementary log-log (clog-log) transformation

Transform: $\log\{-\log \hat{S}(t)\}$ — this lives in $(-\infty, +\infty)$, so a normal CI always works.

Compute CI on the transformed scale, then back-transform.

This is what R's `survfit()` uses by default.

In practice: R computes CIs automatically. Use `ggsurvplot()` to plot them.

Example 2.5 — IUD Data: SE and 95% CI

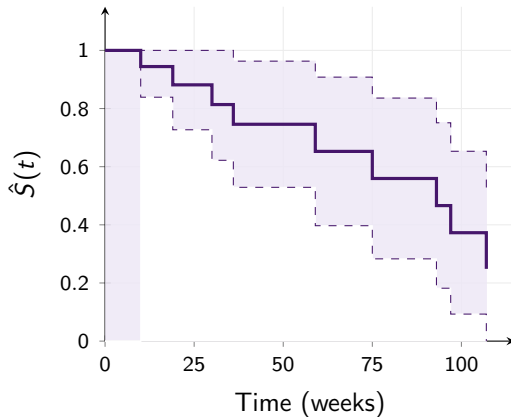
Time	$\hat{S}(t)$	$\widehat{\text{se}}\{\hat{S}(t)\}$	95% CI
0–	1.0000	0.0000	—
10–	0.9444	0.0540	(0.839, 1.000)
19–	0.8815	0.0790	(0.727, 1.000)
30–	0.8137	0.0978	(0.622, 1.000)
36–	0.7459	0.1107	(0.529, 0.963)
59–	0.6526	0.1303	(0.397, 0.908)
75–	0.5594	0.1412	(0.283, 0.836)
93–	0.4662	0.1452	(0.182, 0.751)
97–	0.3729	0.1430	(0.093, 0.653)
107	0.2486	0.1392	(0.000, 0.522)

Limits outside (0, 1) replaced by 0 or 1.

Example 2.5 — IUD Data: KM Curve with Confidence Bands

What to notice:

- ▶ The **shaded band** shows where the true $S(t)$ plausibly lies
- ▶ Early on, the band is narrow (many patients at risk)
- ▶ Later, the band **widens** dramatically (few patients remain)
- ▶ At $t = 107$: CI is $(0.00, 0.52)$ — very uncertain



Key insight

The CI is **pointwise**: it applies to one specific t , not the whole curve.

From $\hat{S}(t)$ to $\hat{H}(t)$

Recall from Chapter 1: $H(t) = -\log S(t)$

Once we have $\hat{S}(t)$, we get $\hat{H}(t)$ immediately.

From KM estimate

$$\hat{H}(t) = -\log \hat{S}(t)$$

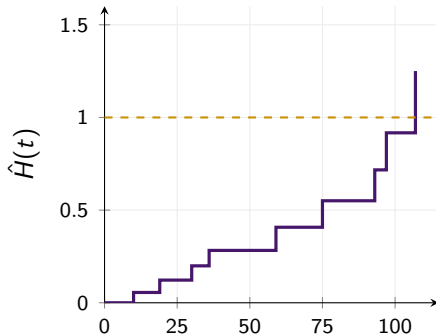
Undefined when $\hat{S}(t) = 0$.

From Nelson-Aalen

$$\tilde{H}(t) = \sum_{j: t_{(j)} \leq t} \frac{d_j}{n_j}$$

Cumulative death probabilities. Always defined.

Nelson-Aalen $\hat{H}(t)$ — IUD data



Interpreting $\hat{H}(t)$

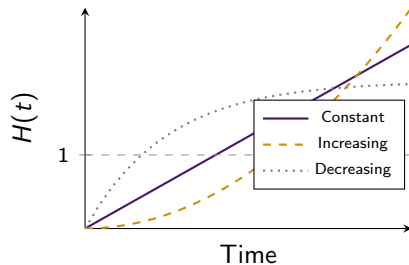
What does $H(t) > 1$ mean?

$H(t)$ can exceed 1 — it is **not** a probability!

Interpretation: expected number of events in $(0, t)$.

When $H(t) = 1$: on average, each person “should have had” 1 event by time t .

- ▶ When $H(t) = 1$: $S(t) = e^{-1} \approx 0.37$
- ▶ So $S(t) < 0.37$ whenever $H(t) > 1$
- ▶ A **linear** $H(t)$: suggests constant hazard (exponential distribution)
- ▶ A **steepening** $H(t)$: hazard increasing over time (things getting worse)
- ▶ A **flattening** $H(t)$: hazard decreasing (survivors becoming more robust)



Estimating the Hazard Function $h(t)$

Kaplan-Meier type estimate

$$\hat{h}(t) = \frac{d_j}{n_j \tau_j}, \quad t_{(j)} \leq t < t_{(j+1)}$$

where $\tau_j = t_{(j+1)} - t_{(j)}$ is the **length of the interval**.

This gives the **number of deaths per unit time** in interval j .

Problem: the estimate is very spiky

Each interval has only 1 death (when $d_j = 1$).

The length τ_j varies a lot (e.g. 4 weeks vs. 23 weeks).

So $\hat{h}(t)$ jumps up and down, making patterns hard to see.

Smoothing the Hazard Function

Kernel smoothing — the idea

At any time t , take a **weighted average** of the estimated hazard values at nearby death times. Death times *close to* t get more weight; those far away get less weight.

The width of the neighbourhood is controlled by the **bandwidth** b .

- ▶ **Small** b : little smoothing, noisy, lots of detail
- ▶ **Large** b : heavy smoothing, cleaner pattern, less detail
- ▶ Optimal b balances bias and variance

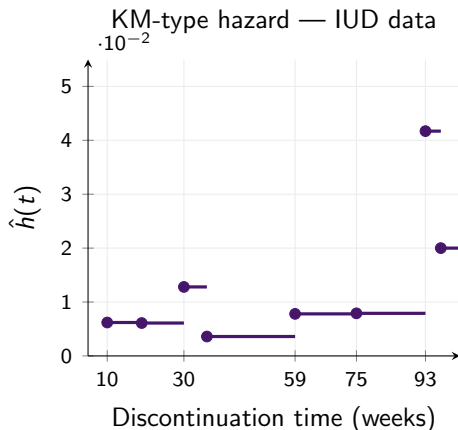
In this course

We focus on **model-based** estimates of $h(t)$ (Chapter 5), which are more interpretable and useful than the non-parametric smoother.

Hazard Function — IUD Example

$t_{(j)}$	τ_j	n_j	d_j	$\hat{h}$
10	9	18	1	0.0062
19	11	15	1	0.0061
30	6	13	1	0.0128
36	23	12	1	0.0036
59	16	8	1	0.0078
75	18	7	1	0.0079
93	4	6	1	0.0417
97	10	5	1	0.0200

Note: $\hat{h}$ at $t = 93$ is high because $\tau_j = 4$ is very short.



Why the Median, Not the Mean?

Problem with the mean

To compute the mean survival time, you need to observe *all* events.

But many observations are **censored** — we never see their event.

Even if everyone eventually experienced the event, survival times are **skewed right**: a few long survivors pull the mean up.

Why the median works

The median is the time at which $S(t) = 0.5$.

We just need the KM curve to cross the 50% line.

It is not affected by very long survival times.

It is the **preferred** summary measure for survival data.

The median survival time is the time by which **half** of the population is expected to have experienced the event.

Finding the Median from the KM Curve

Definition

The estimated median is the **smallest death time** at which $\hat{S}(t)$ drops below 0.5:

$$\hat{t}_{(50)} = \min\{t_{(j)} \mid \hat{S}(t_{(j)}) < 0.5\}$$

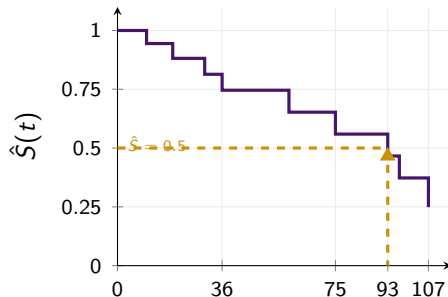
IUD example

From Table 2.2:

$\hat{S}(75) = 0.5594 > 0.5$ (above the line)

$\hat{S}(93) = 0.4662 < 0.5$ (below the line)

$\Rightarrow \hat{t}_{(50)} = \mathbf{93 \text{ weeks}}$



Other Percentiles and the SIQR

p th percentile of survival times

The p th percentile $t_{(p)}$ is the time by which fraction $p/100$ of the population has experienced the event.

Equivalently: $S\{t_{(p)}\} = 1 - p/100$.

Estimated by: smallest $t_{(j)}$ with $\hat{S}(t_{(j)}) < 1 - p/100$.

Semi-Interquartile Range (SIQR)

$$\text{SIQR} = \frac{t_{(75)} - t_{(25)}}{2}$$

Measures the spread of the survival time distribution. The larger the SIQR, the more dispersed the survival times.

Confidence Interval for the Median (1/2)

The idea

Our estimated median $\hat{t}_{(50)} = 93$ weeks is based on a sample.

A different sample might give a slightly different answer.

We want to provide an interval: plausible values for the *true* median.

Standard error via the delta method

$$\widehat{\text{se}}\{\hat{t}_{(50)}\} = \frac{1}{\hat{f}\{\hat{t}_{(50)}\}} \widehat{\text{se}}[\hat{S}\{\hat{t}_{(50)}\}]$$

where $\hat{f}$ is the estimated probability density at the median, obtained from the *slope* of $\hat{S}(t)$ near the median:

$$\hat{f}\{\hat{t}_{(50)}\} = \frac{\hat{S}(\hat{u}_{(50)}) - \hat{S}(\hat{l}_{(50)})}{\hat{l}_{(50)} - \hat{u}_{(50)}}$$

Confidence Interval for the Median (2/2)

Finding $\hat{u}_{(50)}$ and $\hat{l}_{(50)}$

- ▶ $\hat{u}_{(50)}$ = largest death time with $\hat{S}(t) \geq 0.55$
- ▶ $\hat{l}_{(50)}$ = smallest death time with $\hat{S}(t) \leq 0.45$

These are the death times just *above* and just *below* the 50% line.

Example 2.10 — IUD data

From Table 2.4:

$$\hat{u}_{(50)} = 75 \text{ (since } \hat{S}(75) = 0.5594 \geq 0.55) \quad \hat{l}_{(50)} = 97 \text{ (since } \hat{S}(97) = 0.3729 \leq 0.45)$$

$$\hat{f} = \frac{0.5594 - 0.3729}{97 - 75} = \frac{0.1865}{22} = 0.0085 \quad \widehat{\text{se}}\{\hat{t}_{(50)}\} = \frac{0.1452}{0.0085} = 17.1 \text{ weeks}$$

95% CI: $93 \pm 1.96 \times 17.1 = (59, 127)$ weeks

Questions?

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